Since p can be moved before w', it cannot happen-after any event in w; since p occurs the after w, it cannot happen-before any event in w, hence $E \vdash p \diamond w$. Since p can be performed directly after E, we have $p \in enabled(s_{[E]})$.
(Reverse direction) If $p \in I_{[E]}(w)$, then $p \in WI_{[E]}(w)$ follows directly (letting v be the empty sequence). If $p \in enabled(s_{[E]})$ and $E \vdash p \diamond w$, then p is not related with w in the happens-before relation, hence $E.w.p \simeq E.p.w$, implying $p \in WI_{[E]}(w)$.

□

Definition 4.3 (Source Sets). Let E be an execution sequence, and let W be a set of sequences w, such that $E \vdash w$ for each $w \in W$. A set P of processes is *a source set for W after E* if for each $w \in W$ we have $WI_{[E]}(w) \cap P \neq \emptyset$.

The key property is that if P is a source set for W after E, then for each execution sequence of form $E.w$ with $w \in W$, there is a process $p \in P$ and a sequence w' such that $E.p.w' \simeq E.w.v$ for some sequence v. Therefore, when an exploration algorithm intends to cover all suffixes in W after E, the set of processes that are chosen for exploration from $s_{[E]}$ must be a source set for W after E. We formulate this observation as a theorem.

Theorem 4.4 (Key Property of Source Sets). *Let E be an execution sequence, let W be a set of continuations of E, and let W' be a subset of W such that for each $w \in W$ there is a $w' \in W'$ with $E.w' \simeq E.w.v$ for some v. Then the set of first processes of sequences in W' is a source set for W after E.*

This theorem implies that a necessary condition for the correctness of *any* DPOR algorithm is that the set of explored process steps is a source set. It it also a sufficient condition: the proofs of correctness for our algorithms source-DPOR (Theorem 5.2) and optimal-DPOR (Theorem 7.4) contain as key lemmas the property that the set of explored processes is a source set (Claim 5.4 and Claim 7.5, respectively).

Before continuing, we first generalize the relations $p \in I_{[E]}(w)$ and $p \in WI_{[E]}(w)$ to the case where p is a sequence. These generalizations will be used in the proof of soundness of the source-DPOR algorithm, and in the definition of wakeup trees.

Definition 4.5. Let E be an execution sequence and let v and w be sequences of processes.

- Let $v \sqsubseteq_{[E]} w$ denote that there is a sequence v' such that $E.v.v'$ and $E.w$ are execution sequences with $E.v.v' \simeq E.w$. Intuitively, $v \sqsubseteq_{[E]} w$ if, after E, the sequence v is a possible way to start an execution that is equivalent to w.
- Let $v \sim_{[E]} w$ denote that there are sequences v' and w' such that $E.v.v'$ and $E.w.w'$ are execution sequences with $E.v.v' \simeq E.w.w'$. Intuitively, $v \sim_{[E]} w$ if, after E, the sequence v is a possible way to start an execution that is equivalent to an execution sequence of form $E.w.w'$.

It can be seen that Definition 4.5 generalizes Definition 4.1, since for a process p we have $p \in I_{[E]}(w)$ iff $p \sqsubseteq_{[E]} w$, and $p \in WI_{[E]}(w)$ iff $p \sim_{[E]} w$.

As examples, in Fig. 4, we have $q.r \sqsubseteq_{[E']} q.q.r.r$ but $q.q \not\sqsubseteq_{[]} E'r.r$. We also have $q.q \sim_{[E']} r.r$ since $E'.q.q.r.r \simeq E'.r.r.q.q$. Note that $\sim_{[E]}$ is not transitive. The relation $v \sim_{[E]} w$ can be checked using the properties specified in the following lemma.

Lemma 4.6. *The relation $v \sim_{[E]} w$ holds if either*

(1) $v = \langle\rangle$, or
(2) v is of form $p.v'$, and either
  (a) $p \in I_{[E]}(w)$ ("p is an initial of w after E") and $v' \sim_{[E.p]} (w \setminus p)$, or